

Materials and manufacturing of advanced industrial gas turbine components

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Abstract

Industrial gas turbines (IGTs) require materials serving 50 000 h at temperatures up to 1050°C under high loading in an aggressive environment. These requirements are different compared to the aircraft engines, which have typically higher peak temperature for a short time, but with cruise cycle temperature significantly lower than the base load temperature for IGT. Gas turbine material development and application is always a compromise between strength and stability, between oxidation and hot corrosion resistance, between coating protection and diffusion-induced degradation, between application of most advanced materials and manufacturing technologies and requirement for an extensive engine experience. This paper gives an overview of gas turbine materials selection, trends in high-temperature materials, environmental and temperature-protective coatings, and in some of manufacturing techniques for industrial gas turbine components. © 2001 Elsevier Science B.V. All rights reserved.

Keywords: Industrial gas turbines; Single crystals; Superalloys; Casting; Oxidation protection; Coatings

1. Introduction

The gas turbine inlet temperature in modern industrial turbines exceeds 1300°C, the life requirement for the highly loaded high-temperature components is 25 000–50 000 h, and efficiency requirements limit the amount of compressor air available for cooling of turbine parts.

Traditionally, material technology for power generation gas turbines has followed the aircraft technology, being 10–20 years behind [1]. During the last decade, dramatic competition in the power equipment industry has boosted technology to the level achieved in the aviation turbines just years before (Fig. 1). Latest industrial gas turbines (IGTs) use single crystal and directionally solidified blades and vanes, rhenium-contained nickel-based superalloys, advanced metallic and ceramic coatings. The large size of IGT components shows limits of traditional casting techniques, and newly developed (gas cooling casting) or rediscovered (liquid metal cooling) processes are being extensively evaluated. Single crystal components require also a new generation of reconditioning technologies, such as epitaxial brazing and laser cladding. Differences in component size and, especially, in operation conditions, will result in stronger separation of aero- and IGT material development lines in future.

2. Single crystal alloys

Advanced IGT blading is cast in single crystal Ni-based superalloys, mostly Re-containing. The future trend in IGT alloy development reflects the specific power generation requirements: IGT components serve at base load temperature much longer than those in aircraft engines, though the peak temperature is lower. Fig. 2 illustrates an approach, that for further extension of IGT blading material life, alloys with lower precipitate volume fraction but more stable γ – γ' structure (low lattice mismatch) have better potential. γ/γ' lattice mismatch causes additional strengthening, and this is widely used in commercial second and third generation superalloys. However, the lattice mismatch as well as very high γ' volume fraction promote formation of coarse or rafted structure [2], especially detrimental when γ' becomes a continuous phase. Fig. 3 illustrates effect of creep strain on γ' morphology at high temperature. Though in some works the γ/γ' inversion was reported to show no detrimental effect on creep properties [3], ductility and TMF life definitely deteriorate. Alloys with higher resistance to rafting and γ/γ' inversion, namely those ones with moderate γ' volume fraction and low lattice mismatch fit better to IGT requirements, especially if their strength will be improved compared with commercial low γ' volume alloys such as CMSX 11 [4].

Rejuvenation of the γ' structure in highly loaded SX blades has an additional limitation in recrystallisation due to accumulated strain. To prevent recrystallisation,

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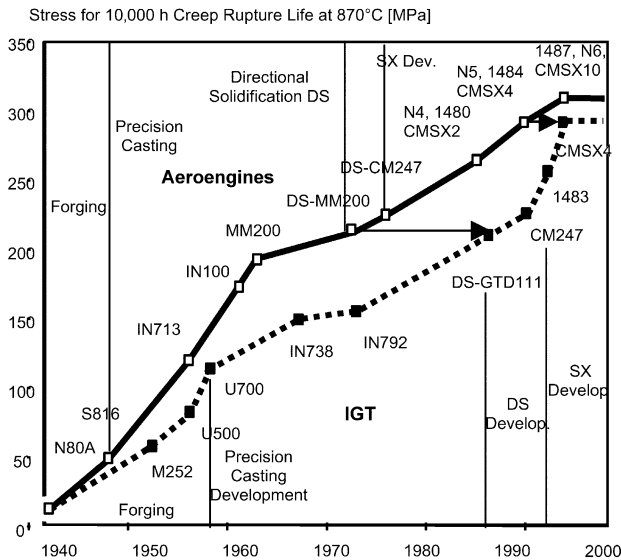


Fig. 1. Progress in aero- and IGT blade material.

the maximum rejuvenation heat treatment cycle temperature has to be limited below γ' solvus temperature. Fortunately, the rafted structure is energetically highly unstable and therefore restoration of the cubical γ' morphology is possible without full solutioning (Fig. 3). Resulting structure, though coarser than the initial one, provides extension of the component life by 50% and more.

Another requirement to IGT blade materials is improved oxidation resistance, promoted by clean processing, desulfurisation and addition of rare-earth elements [5]. Interestingly, that at 950°C addition of Y has not shown a positive effect confirmed for higher temperatures (Fig. 4). Longer test is needed to make a final conclusion on role of Y at

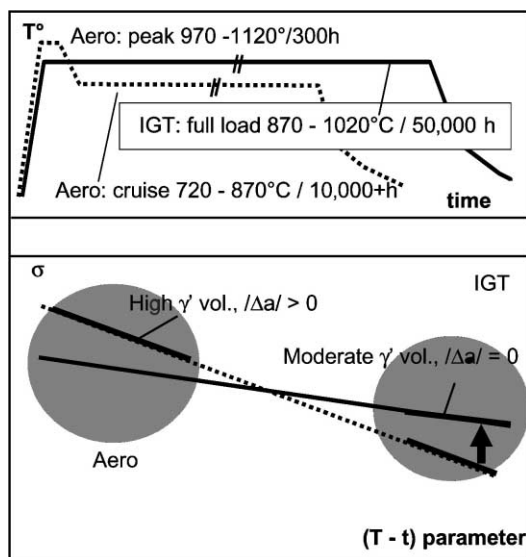
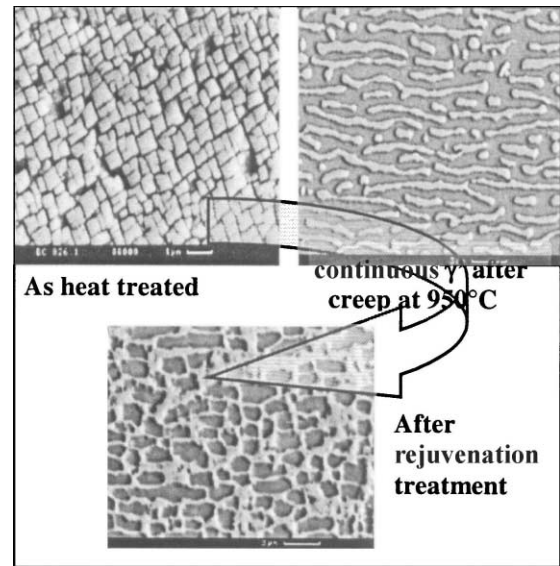


Fig. 2. Specific IGT service conditions and trends in blade alloy development.

Fig. 3. Transition in γ' morphology during creep of CMSX-4 at 950°C.

moderate temperatures. The next generation of IGT blade alloys will combine the high oxidation and fatigue properties of aircraft materials with a significantly increased resistance to structure degradation.

With regard to manufacturing, large IGT single crystal components cause very high costs using traditional Bridgman casting technology. Fig. 5 illustrates the current Bridgman process. The mould with molten metal moves gradually from the heating chamber to the cooling chamber separated by baffle. Heat is removed by radiation in vacuum from the mould towards water-cooled chamber walls. Good isolation between cold cooling chamber and hot heating chamber provided by baffle results in vertical temperature gradient.

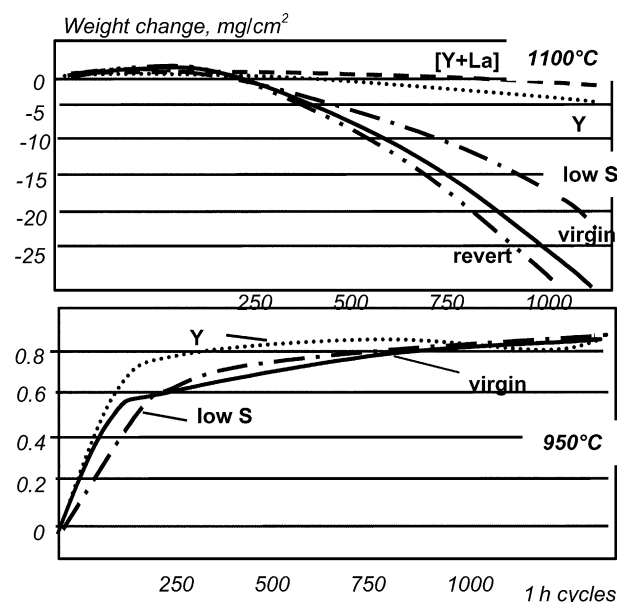


Fig. 4. Influence of alloy cleanliness and rare-earth elements addition on CMSX-4 cyclic oxidation at 1100 and 950°C.

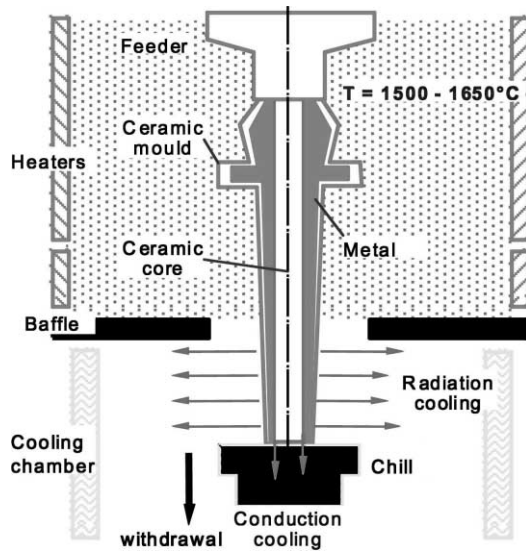
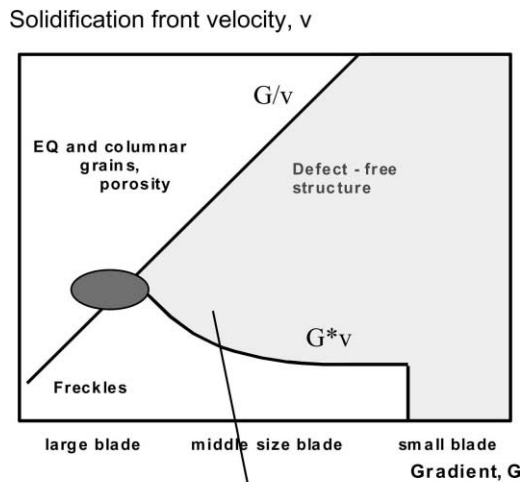


Fig. 5. Bridgman casting process.

High temperature gradient in vertical direction provides directional solidification of the component. For large cluster moulds with massive components this type of heat extraction, however, meets its limits. Schematic casting defect map (Fig. 6) shows limitations in further improvement in casting yield for conventional process. When the temperature gradient is high enough, no grain defects occurs. Heat radiation in vacuum cannot provide significant gradient to massive IGT parts. At low gradient conditions, too high or too low solidification rate result in various casting defects, such as freckles, columnar grains and porosity. Optimisation of the casting process parameters by computer process modelling and design for manufacturing help to increase yield and reduce process development time. More radical is a development of the single crystal casting techniques with high cooling rate, such as liquid metal cooling (LMC) [6] and gas



- Large IGT SX blades: low yield, high cost

Fig. 6. Casting defect map for directional solidification process.

- LAGB Tolerance (GB elements)
- C, B, Hf additions
- low angle grain boundary strengthening

LCF properties

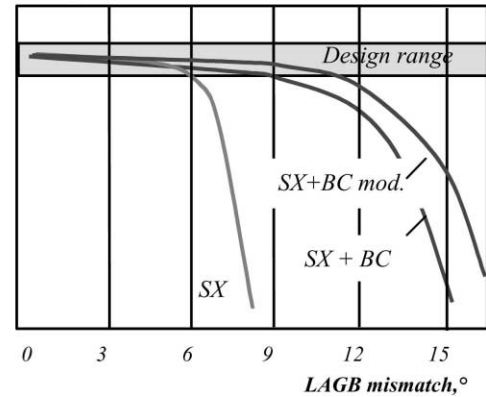
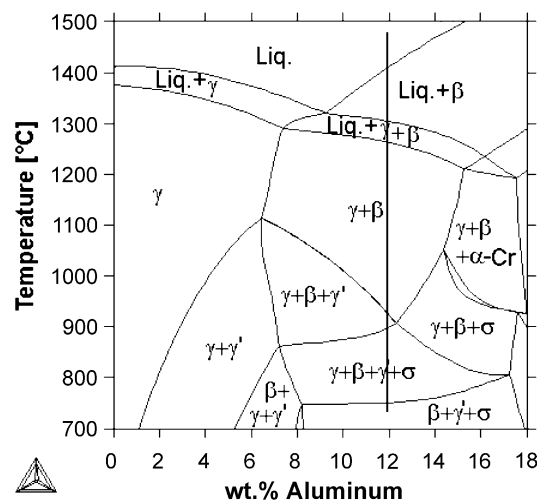


Fig. 7. LAGB tolerance (LCF) of SX alloys.

cooling casting (GCC) process [7]. Alternative casting processes improve yield, but also increase production capacities, primarily by allow casting of more blades per cluster.

Alloy design for castability is another way to avoid many of the casting defects. The density balance between the liquid metal in the mushy zone and the bulk liquid significantly reduce metal convection during solidification. This prevents formation of freckles and increases tolerance to the inclination of solidification front with respect to formation of secondary grains. Balance of elements such as Hf and C controls metal–mould reaction (wrinkles, surface scale, etc.). Additions of grain boundary-active elements (C, B, Hf, Zr) increase tolerance to low angle grain boundaries misfit from 5–6° to 9–12° (Fig. 7).

Blading materials shall be developed and selected as a system, which includes base material and coating. They

Fig. 8. Phase modelling for coating design, section of Ni–Co–Cr–Al–Si phase diagram. 12% Al provides the most stable γ – β system.

require compatible manufacturing process parameters and optimisation of interdiffusion zone. Modelling programs, such as ThermoCalc and DICTRA, allow fast and effective pre-selection of potential compositions (Fig. 8).

3. Gas turbine coatings

A great variety of coating compositions and technologies are used in IGTs. On first stages most of IGT manufacturers use vacuum plasma-sprayed (VPS) MCrAlY coatings with or without air plasma sprayed ceramic thermal barrier coating (TBC) as a best compromise of temperature capability, life and costs. EB PVD ceramic coatings, though provide excellent stress tolerance and smooth aerodynamically beneficial surface, are significantly more expensive. Last stages and structural parts usually coated with oxidation or corrosion-protective metallic systems using lower cost processes (HVOF, chromizing, paintcoat).

Oxidation protection remains the most important function of metallic coatings. In base load machines, coatings must survive more than 20 000 h at temperatures close to 1000°C. Coating life is determined by amount of oxide builder (Al for Al_2O_3 forming coatings), its activity and adhesion of oxide scale [8]. Adhesion can be improved by addition of surface-active elements (Y, La, Ce, Hf, etc.) and optimisation of oxide morphology, for instance, by heat treatment in controlled atmosphere. The Al-reservoir is determined by coating thickness, Al content, phase structure (activity of Al in the system) and by control of interdiffusion process. On the other hand, increased coating thickness, Al-rich and especially high Al-activity chromium-rich γ - β - α (σ) systems have reduced mechanical properties. Generally, detrimental for coating mechanical life is also diffusion-protective γ' layer on the coating-base metal interface.

Advanced IGT coatings do not reduce fatigue life of the base alloy [9] (Fig. 9). This is true, however, only until

coating layer exceed a certain thickness. Manufacturing control over coating thickness along the components plays therefore a key role in component life. Creep resistance of the coating is important in blade tip area. Customised phase structures, such as a balance of Al, Cr-rich but brittle reservoirs with ductile phases, is a general approach in coating design. Alloying of β -phase can additionally improve coating ductility and toughness.

Despite high gas and metal temperatures in advanced turbines, corrosion still must be seriously considered in hot path material selection, especially for oil-operated and dual fuel engines. Significantly increased pressure on first stages enables deposition of corrosion products at surface temperatures above 850°C. Since alloying elements influence oxidation and corrosion resistance differently, it is important either to have a choice of coatings for various applications or to use a coating with compromised properties. Corrosion model, which considers S, K and Na content in turbine environment, temperature and pressure (Fig. 10) allows the right coating selection for each turbine stage.

The most critical area in high-temperature coating application currently seems to be, however, not the coating chemistry but manufacturing process control. Out-of-spec coating thickness, porosity and interface quality result in much faster coating failure than an insufficient oxidation or corrosion resistance. The trend is to optimise stability of current processes as well as to introduce new technologies. Plasma spray process modelling supports to improve quality and reduce costs of coated components. Advances in HVOF and in arc deposition technologies allow to extend their use to high-temperature IGT components instead for LPPS [10]. High thermo-mechanical loading of advanced single crystal blades promotes development of new types of coatings, such as epitaxial single crystal MCrAlY coating [11]. Columnar, directionally pre-cracked TBC structures provide better spallation resistance than conventionally air plasma-sprayed TBCs at costs lower than those of EB PVD process. Tailored

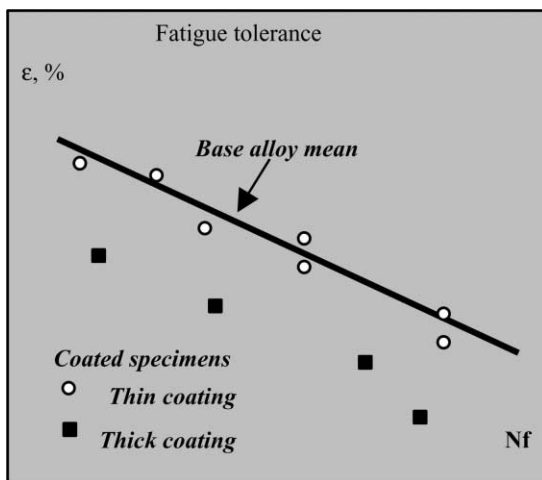


Fig. 9. Fatigue properties of coated CMSX-4 samples, coated with VPS MCrAlY coating, 1050°C.

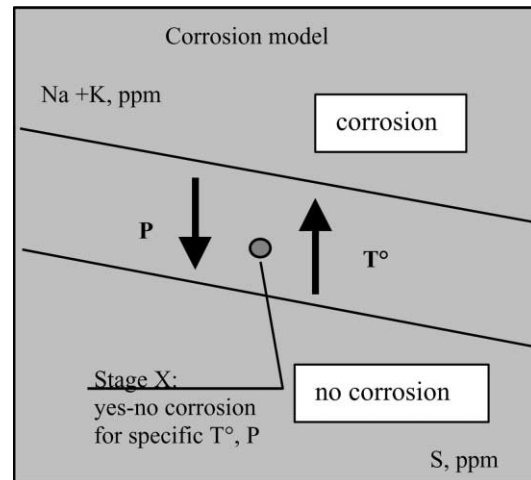


Fig. 10. Model determines corrosion conditions at turbine stage for P and T° (Na, K and S content).

porosity morphology can significantly increase temperature resistance of yttria-stabilised zirconia (YSZ) ceramic. We believe that the next sizeable step in coating area will be provided by new manufacturing processes rather than new coating systems.

4. Outlook

The question, what will replace single crystal Ni-base superalloys is discussed at least last 25 years with always the same potential candidates if some setbacks are solved.

High temperature Si-based ceramics (nitrides and carbides) have been used in turbine projects in USA and Japan [12,13], and demonstrated certain potential for small turbines. In situ reinforced microstructures give improved toughness and reliability compared to the first generations of monolithic ceramics. Still, a foreign object impact tolerance (FOD), oxidation protection at temperatures above 1200°C, especially in water vapour containing environment remains the main hurdles for IGT application. Manufacturing, though well advanced during last decade, is very time- and cost-intensive. Tests in larger gas turbines have not been successful yet. Ceramic parts require special design and probability-based philosophy in life time assessment methodology.

Ceramic matrix composites (CMC) have been recently tested in combustor liner application [14]. Material and functional failures indicate that problems are similar to those of monolithic ceramic, though toughness of CMC is much higher. CMC require protection of fibres and time-intensive infiltration technique and are prohibitively expensive for large-size IGT applications.

The new generation of high-temperature metallic systems, Ir- and Rh-based “refractory superalloys” [15] requires further investigation of mechanical and environmental properties and development of manufacturing technology and equipment for component production at much higher temperatures than those currently used.

Most probably, the next generation in IGT material application comes not from new material systems, but from alternative manufacturing technologies, providing better quality, new properties level through new structures and

lower costs. New challenge to materials and manufacturing technologies is also to provide solutions for advance design concepts, such as supercooled blading. New core materials are used for casting of components with superfine wall internal cooling geometry. Competitive to the casting process are various build-up techniques, such as epitaxial cladding of outside blade surface over pre-cast component with open wall cooling structure. Materials and coatings for supercooled applications require much higher tolerance to line defects and thermal stresses.

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